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A Theorem and Proof

Definition 1. $g(\lambda) := \mathbb{KL}(p_h || (1 + \lambda)p'_h - \lambda p_m)$.

Proposition 2. $g(\lambda)$ is a convex function.

Proof. $\mathbb{KL}$ is convex and $g(\lambda)$ is the restriction of $\mathbb{KL}$ on a line, so $g(\lambda)$ is also convex. $\square$

Proposition 3. If $g(\lambda)$ is not constant, then $g(\lambda)$ has a unique minimum point.

Proof. The non-constancy of $g(\lambda)$ implies that $p'_h \neq p_m$. The domain of $g(\lambda)$ is

$$I := \bigcap_{v \in \mathcal{V}} \{\lambda \in \mathbb{R} : 0 \leq (1 + \lambda)p_h^{(v)} - \lambda p_m^{(v)} \leq 1\}.$$

$[-1, 0] \subseteq I$ so I is non-empty. I is the intersection of some closed intervals, so I is also a closed interval. Note that $g(\lambda)$ is continuous, so $g(\lambda)$ has a minimum value.

Assume that $g(\lambda)$ has minimum points λ_1 and λ_2 with $\lambda_1 < \lambda_2$. By the convexity, $g(\lambda)$ is constant on $[\lambda_1, \lambda_2]$. Note that $g(\lambda)$ is an analytic function. By Corollary 1.2.6 in (Krantz and Parks, 2002), the constancy of $g(\lambda)$ on $[\lambda_1, \lambda_2]$ implies the constancy on I . This contradicts that $g(\lambda)$ is not constant, so the minimum point of $g(\lambda)$ is unique. $\square$

Remark. Corollary 1.2.6 in (Krantz and Parks, 2002) applies to open intervals, and $g(\lambda)$ is continuous at the endpoints, so it also applies to our closed intervals, $[\lambda_1, \lambda_2]$ and I .

Definition 2. The minimum point of $g(\lambda)$ is λ_* .

Theorem 1. If $g'(0) < 0$, then $\lambda_* > 0$ and for any $\lambda \in (0, \lambda_*]$, we have

$$\mathbb{KL}(p_h || (1 + \lambda)p'_h - \lambda p_m) < \mathbb{KL}(p_h || p'_h). \quad (7)$$

Proof. $g'(0) < 0$ implies that $g(\lambda)$ is not constant, so λ_* is well-defined by Proposition 3.

$g'(0) \neq 0$ implies $\lambda_* \neq 0$. Assume $\lambda_* < 0$. By the first-order condition of convex functions,

$$g(\lambda_*) \geq g(0) + g'(0)\lambda_* > g(0).$$

This contradicts that $g(\lambda_*)$ is minimum, so $\lambda_* > 0$.

By the definition of convex functions, we have

$$\begin{aligned} g(\lambda) &\leq \frac{\lambda}{\lambda_*} g(\lambda_*) + \left(1 - \frac{\lambda}{\lambda_*}\right) g(0), \\ &= g(0) - \frac{\lambda}{\lambda_*} (g(0) - g(\lambda_*)), \\ &< g(0) \end{aligned}$$

for any $\lambda \in (0, \lambda_*]$. $g(\lambda) < g(0)$ is just (7). $\square$

B Experimental Details

We follow the implementation of (Wang et al., 2024; Bao et al., 2024) to generate machine texts for XSum and SQuAD while adopting the approach of Dipper (Krishna et al., 2023) for LongQA. To reproduce DetectGPT and Fast-DetectGPT, we align with (Wang et al., 2024) and employ GPT2-XL (Radford et al., 2019) as the surrogate model to generate samples. As for DNA-GPT (Yang et al., 2024), we adopt the basic setup surrogated on GPT-3.5-turbo. For the LLM decoding, our experiments adopt the default setup of sampling parameters, *i.e.*, no probability clipping and $T = 1$. We run all experiments in NVIDIA RTX A6000 GPUs.

Below, we provide our carefully designed prompts to elicit human-like and machine-like token distributions from the off-the-shelf LLM.

(1) For human-style distributions: An intuitive way is to construct prompts that directly ask LLMs to produce human-like sentences by encouraging vivid lexical substitution and diverse sentence structures. However, compared to a direct request for human-toned texts, we empirically find that framing a realistic human conversation scene to LLMs leads to more vivid and diverse generations, strengthening the attack effectiveness.

System Prompt

You are a helpful paraphraser. You are given an input passage 'INPUT'. You should paraphrase 'INPUT' to print 'OUTPUT'. 'OUTPUT' should preserve the meaning and content of 'INPUT'. 'OUTPUT' should not be very shorter than 'INPUT'.

This system prompt is similar to that in (Sadasi-van et al., 2023) and instructs the LLM to act as a paraphraser while maintaining the text quality of the sentences before the paraphrasing.

User Prompt

Rewrite the following INPUT in the tone of a text message to a friend without any greetings or emojis:

Rather than making a direct request for human-toned texts, we experimentally find that presenting a realistic conversation scene with humans can better guide the LLM to produce more vivid and diverse sentences, further boosting stronger attacks.